

Engression on Real ENA Weather Data: A Brief Summary

WeatherEngression Research Notes

August 1, 2026

1 Model Architectures

Every model is just a sampler $\hat{Y} = g(X, \varepsilon)$: feed in the weather X plus a shot of random noise ε and you get one possible CCN; draw many times and you trace out the whole distribution $P(Y | X)$. All are trained the same way (the energy-score / CRPS loss). The only thing that changes between them is how X is read and where the noise goes in.

- **Built-in (vanilla).** The public **engression** model: X is flattened and fed to a StoNet (MLP with fresh noise concatenated at every layer). No time order.
- **LSTM + default head.** LSTM encoder $\rightarrow$ summary $h(X)$; one noise vector concatenated to $h(X)$, then a plain MLP. $\hat{Y} = g(h(X), \varepsilon)$.
- **LSTM + StoNet head.** Same encoder; the head re-injects noise at every layer. No monotonicity.
- **LSTM + pre-additive head.** $Y = g(\phi(X) + \eta)$: the encoder makes a low-dimensional index $\phi(X)$, noise η is *added*, and a monotone g maps to the target. The only head with the paper's extrapolation theory.

In every case the noise enters late, after a deterministic encoder — the randomness lives in the final CCN, and noise pushed deep into a 241-step recurrence tends to get washed out.

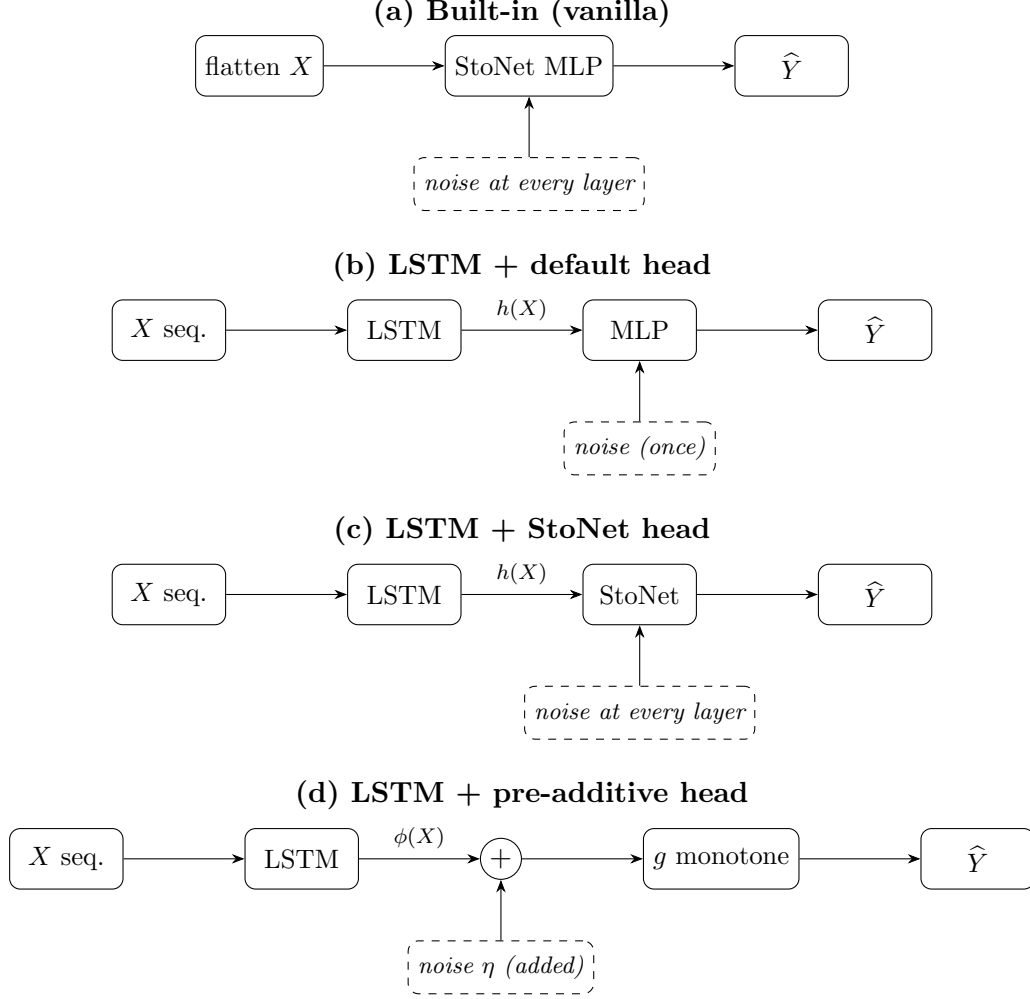


Figure 1: Data and noise inputs per model. X enters from the left, noise from below (dashed). Noise is concatenated in (a)–(c) and added to $\phi(X)$ in (d).

Table 1: Engression assumptions by architecture. *achieved* = holds by construction; *hoped* = the structure is present but the guarantee is not exact; *violated* = the assumption does not hold.

Architecture	Learns $P(Y X)$	Pre-additive noise	Monotone g	Extrapolation theory
Vanilla (flat StoNet)	achieved [†]	violated	violated	none
LSTM + default head	achieved [†]	violated	violated	none
LSTM + StoNet head	achieved [†]	violated	violated	none
LSTM + pre-additive	achieved [†]	achieved	achieved	hoped [‡]

[†] targeted by the energy loss; achieved *in practice* only when the noise channel does not collapse (the under-dispersion of Section 3). [‡] the pre-additive form and a monotone g are present, but the index $\phi(X) = W \cdot \text{LSTM}(X)$ is *nonlinear*, whereas the paper’s extrapolation theorem assumes a *linear* index—so the guarantee is hoped for, not proven.

2 Experimental Setup

Data. Eastern North Atlantic (ENA). Each sample is a 10-day airmass back-trajectory — 241 hourly steps $\times$ 22 weather variables — and we predict $\log_{10}(\text{CCN})$, the cloud-particle count when the air arrives. Dropping samples with a missing target leaves $\approx 60,866$ trajectories.

Train/test split (“paper” split). The test set is six alternating months of 2022 (Jan, Mar, May, Jul, Sep, Nov); everything else trains — 57,646 train vs 3,220 test. Using whole-month blocks stops near-identical neighbouring hours from leaking across the split. Since the test months sit between training months, this tests *interpolation*, not extrapolation.

Dimensions. The LSTM keeps the unflattened sequence, $X \in \mathbb{R}^{57646 \times 241 \times 22}$ (train). The flat (vanilla) model instead flattens each trajectory to $241 \times 22 = 5302$ features.

3 In-Sample Performance

We grade each predicted distribution four ways: **coverage** (does the 90% interval actually hold 90% of the points? closer to 0.90 is better), **width** (narrower = sharper), median absolute error (**MAE**), and **CRPS** (overall fit, lower better).

Our main model is the **LSTM + StoNet head** (Table 2), with the pre-additive head next to it for comparison (Table 3).

Table 2: **LSTM + StoNet head** (primary model): full 8-configuration sweep, sorted by CRPS. Lower width / MAE / CRPS is better; coverage near 0.90 is better. Starred row ($\star$): recommended configuration (Figures 2, 3).

lr	depth	hidden	cov. 90%	cov. 50%	width	MAE	CRPS
0.003	5	192	0.783	0.378	0.50	0.164	0.120
$\star 0.01$	5	128	0.844	0.438	0.65	0.171	0.123
0.003	5	128	0.573	0.241	0.34	0.176	0.137
0.003	3	128	0.458	0.206	0.26	0.176	0.143
0.003	3	192	0.384	0.158	0.22	0.180	0.151
0.01	3	128	0.907	0.505	0.97	0.212	0.152
0.01	5	192	0.762	0.488	0.79	0.216	0.161
0.01	3	192	0.856	0.310	0.86	0.226	0.166

Table 3: **LSTM + pre-additive head**, for comparison. Depth is set by a separate monotone-net parameter, so `num_layer` had no effect; the four distinct configurations are shown. Uniformly weaker than StoNet.

lr	hidden	cov. 90%	cov. 50%	width	MAE	CRPS
0.003	128	0.386	0.162	0.22	0.186	0.157
0.003	192	0.321	0.133	0.18	0.182	0.157
0.01	128	0.838	0.450	0.86	0.233	0.168
0.01	192	0.511	0.216	0.62	0.388	0.311

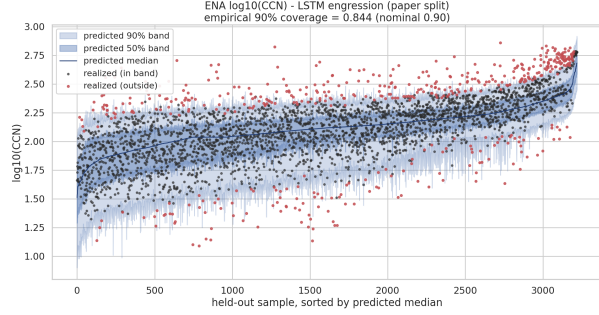
Table 4: **LSTM + default head** (unconstrained MLP). A single full-data run, *not* swept — the baseline referenced below. Best raw CRPS, but lacks the pre-additive structure.

lr	depth	hidden	cov. 90%	cov. 50%	width	MAE	CRPS
0.003	3	192	0.888	0.480	0.69	0.176	0.126

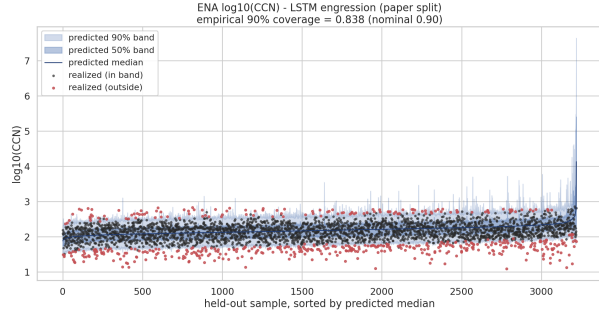
Table 5: **Vanilla (flat) baseline**: full 24-configuration sweep, sorted by CRPS. Starred row (★): the config shown in Figure 2 (highest coverage). The pattern is the flat model’s whole story: the lowest-CRPS configs are sharp but badly *under*-cover (~ 0.3 – 0.5), and the only configs that reach ~ 0.9 coverage (lr 0.01, depth 5) do so with the *widest*, least accurate intervals — it covers only by hedging.

lr	depth	hidden	noise	cov. 90%	cov. 50%	width	MAE	CRPS
0.003	3	256	64	0.392	0.174	0.21	0.167	0.139
0.003	3	256	128	0.416	0.177	0.24	0.173	0.142
0.01	3	128	64	0.536	0.246	0.33	0.183	0.144
0.003	3	128	64	0.426	0.183	0.24	0.177	0.145
0.01	3	256	128	0.539	0.233	0.34	0.189	0.148
0.001	3	256	128	0.304	0.121	0.17	0.172	0.148
0.003	3	128	128	0.398	0.172	0.24	0.181	0.149
0.01	3	256	64	0.421	0.185	0.27	0.184	0.149
0.001	3	256	64	0.310	0.132	0.18	0.174	0.149
0.01	3	128	128	0.532	0.238	0.39	0.195	0.151
0.003	5	128	128	0.379	0.150	0.23	0.183	0.152
0.001	5	256	64	0.268	0.103	0.15	0.175	0.153
0.001	5	256	128	0.263	0.107	0.15	0.177	0.154
0.003	5	256	64	0.273	0.107	0.16	0.179	0.155
0.001	3	128	64	0.289	0.118	0.18	0.180	0.155
0.001	5	128	64	0.290	0.123	0.17	0.179	0.155
0.001	3	128	128	0.287	0.118	0.17	0.180	0.156
0.003	5	256	128	0.287	0.112	0.16	0.180	0.156
0.003	5	128	64	0.311	0.118	0.18	0.184	0.158
0.001	5	128	128	0.295	0.125	0.17	0.182	0.158
★0.01	5	128	64	0.907	0.487	1.02	0.227	0.162
0.01	5	128	128	0.924	0.514	1.09	0.227	0.162
0.01	5	256	64	0.850	0.420	0.84	0.227	0.163
0.01	5	256	128	0.461	0.071	0.53	0.227	0.200

Baselines (context only). The flat *vanilla* model (Table 5) reached coverage 0.907 only by widening (width 1.02, MAE 0.227, CRPS 0.162) — its best CRPS (0.139) comes from a sharp config that under-covers at 0.39. The unconstrained LSTM+MLP head (Table 4) reaches CRPS 0.126, below every vanilla config; we report StoNet as the constrained engression head.



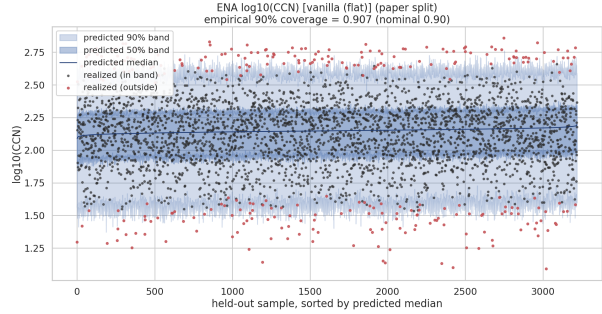
StoNet (5×128): cov. 0.844



Pre-additive (hidden 128): cov. 0.838

default-head chart
pending GPU rerun
(`slurm/regen_default_charts`)

Default head (3×192): cov. 0.888



Vanilla (flat) (5×128): cov. 0.907

Figure 2: In-sample posterior bands for each architecture’s best run. Points sorted by predicted median; shaded bands are the predicted 50% and 90% intervals; dots are realised $\log_{10}(\text{CCN})$ (red = outside the 90% band).

Sweep details. Tables 2–3 are the full LSTM head sweep (16 runs: head $\times$ learning rate $\{0.003, 0.01\} \times$ depth $\{3, 5\} \times$ hidden $\{128, 192\}$, noise dimension fixed at 96). The flat (vanilla) baseline was swept separately over 24 configurations (learning rate $\in \{0.001, 0.003, 0.01\}$, width $\in \{128, 256\}$, depth $\in \{3, 5\}$, noise dimension $\in \{64, 128\}$). Fixed throughout: $\log_{10}(\text{CCN})$ target, paper split, full data, batch size 256, weight decay 0, early stopping on the training energy loss.

Takeaways.

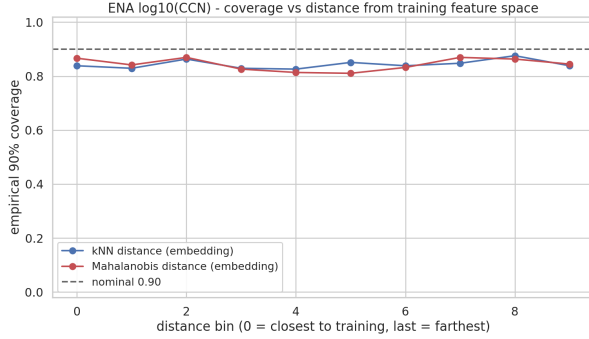
- **Sequence structure helps.** The LSTM is calibrated while staying sharp and accurate; the flat model reaches coverage only with wide, inaccurate intervals.
- **Under-dispersion is the main failure.** Many configurations collapse to a sharp point and lose coverage; coverage *anti-correlates* with sharpness.
- **What controls collapse.** Learning rate dominates (StoNet collapsed at $\text{lr} = 0.003$, recovered at $\text{lr} = 0.01$); more *depth* helps, more *width* hurts.

4 Out-of-Sample Performance

We measure how far each test point sits from the training data (in the LSTM’s embedding space, by k NN and Mahalanobis), then check whether coverage holds up as that distance grows.

Findings.

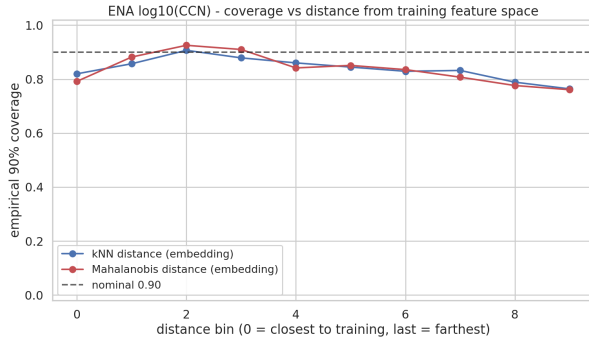
- Both heads’ best runs are roughly *flat* across distance (Figure 3): calibration is robust across the training support. Under-dispersed (collapsed) configurations instead lose coverage with distance.
- k NN and Mahalanobis agree closely.



StoNet (5×128)

default-head chart
pending GPU rerun
(`slurm/regen_default_charts`)

Default head (3×192)



Pre-additive (hidden 128)

Vanilla (flat): N/A
no learned embedding; raw 5302-d distance is degenerate, so this diagnostic does not apply

Vanilla (flat)

Figure 3: Coverage vs. distance from training (in the model’s LSTM embedding space), k NN and Mahalanobis, for each architecture’s best run. The LSTM heads stay roughly flat (paper split is in-distribution). The flat *vanilla* model has no learned embedding, so the diagnostic does not apply.

Caveat. The paper split is in-distribution, so a flat curve is expected and is *not* evidence of extrapolation. An out-of-support split (high-CCN tail `ccn_tail`, or a shifted `event`) is the real test; not yet run.

5 Train/Test Distribution Shift

The paper split is *supposed* to be in-distribution — let’s check. One catch first: adjacent hours are $\sim 99.7\%$ identical, so the 57,646 train / 3,220 test rows are really only a few hundred *independent* episodes. So we thin to every 24th row (2,402 train / 135 test) and describe each trajectory with 132 summary numbers. Wildfire is a similar slice of each side (11.1% train vs 12.7% test), so the split doesn’t lopside the treatment balance.

Three complementary tests ask three different questions.

(1) **Classifier two-sample test (C2ST)** — **are they different, and how?** A classifier trained to tell train-period from test-period trajectories reaches $\text{AUC} = 0.79 \pm 0.02$ (linear probe

0.73), well above the 0.5 coin-flip line, so the periods are clearly distinguishable. The drift is dominated by **carbon monoxide** — a combustion tracer — then pressure and marine/aerosol-source channels (Figure 5).

(2) MMD permutation test — is it statistically real? The maximum-mean-discrepancy statistic sits ~ 7 standard deviations beyond its label-shuffled null: $p = 0.0008$ (the floor for 5000 permutations), matched by an independent energy-distance test ($p = 0.001$; Figure 4a). The shift is significant. (With enough independent episodes *any* nonzero difference becomes significant, so a small p means the shift is *real*, not *large*.)

(3) kNN overlap — does the test still sit inside training territory? Median nearest-neighbour distances are nearly equal (train $\rightarrow$ train 8.01 vs test $\rightarrow$ train 8.36, ratio 1.04), and only 7.4% of test episodes fall beyond the training 95th percentile (Figure 4b). Despite the significant shift, test episodes overwhelmingly land *inside* the training support.

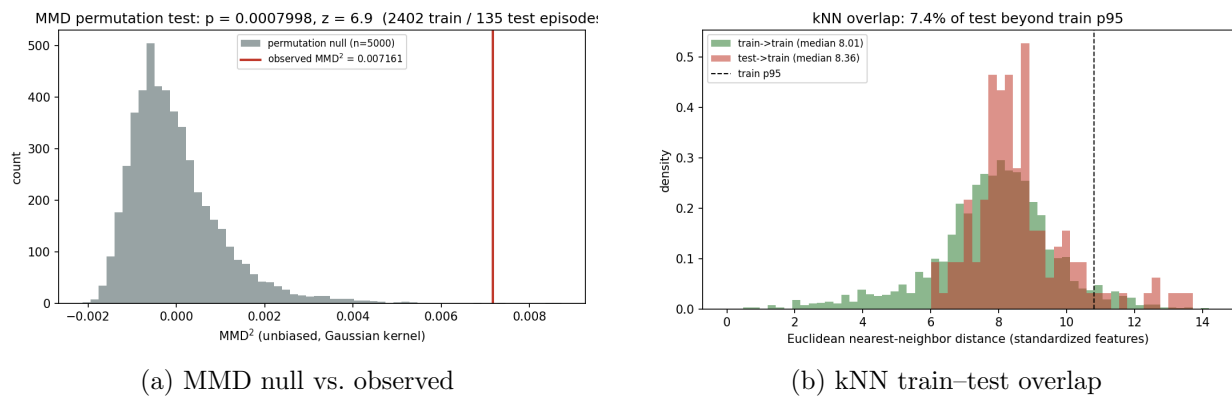


Figure 4: (a) Observed MMD^2 (red) lies far right of the label-shuffled null (grey): the shift is significant. (b) Distance to the nearest training episode; test (red) and train (green) overlap heavily, only a thin tail past the training 95th percentile.

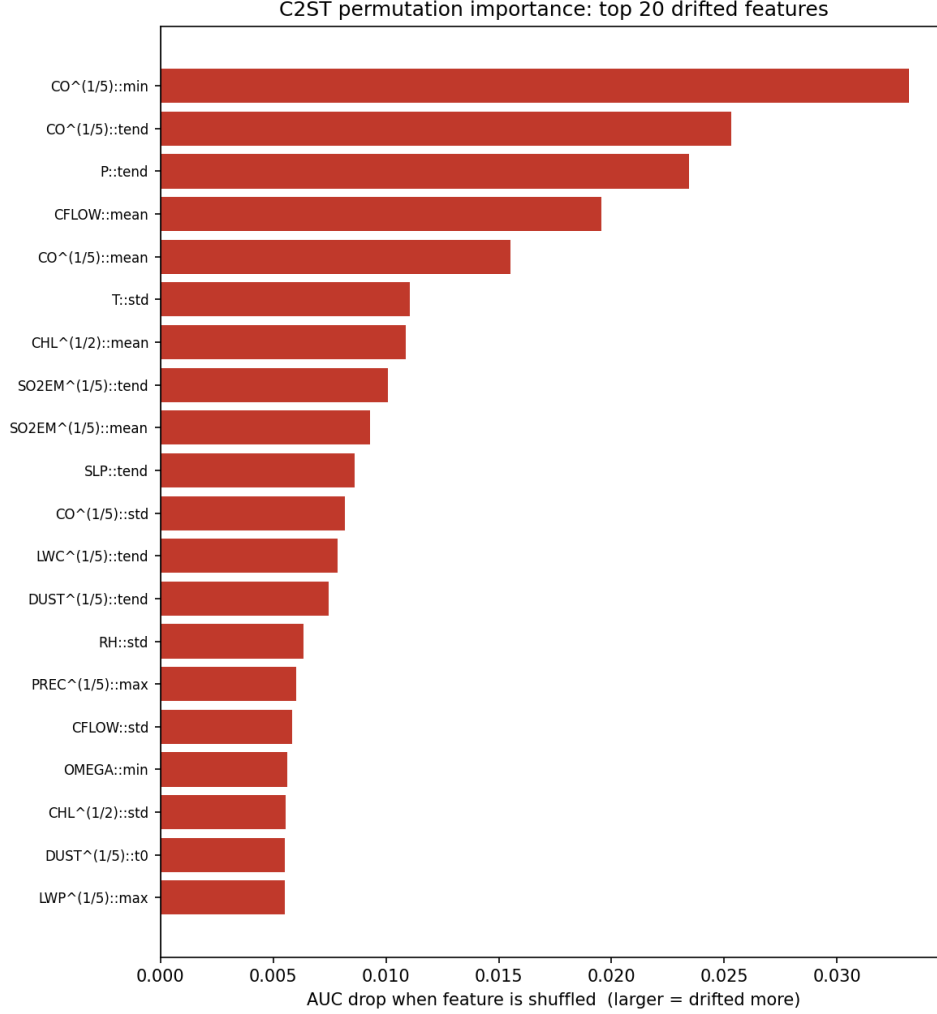


Figure 5: C2ST permutation importance: which channels the classifier used to separate train from test. CO summaries dominate the drift.

Table 6: Distribution-shift verdict. The test-period weather is genuinely and significantly different (a CO / air-mass-source drift), but the difference is modest and stays within the training support — so overlap holds.

Test	Question	Answer
C2ST	Are train and test different?	Yes — AUC 0.79; driven by CO
MMD	Is that difference significant?	Yes — $p = 0.0008$, $\sim 7\sigma$
kNN	Does test stay in training support?	Mostly — ratio 1.04, 7.4% past p95

Why this matters. Since the test still sits *inside* the training data, the effect measured where both models have support — the ATT at wildfire weather (Section 8) — is on solid ground. An ATE/ATC would instead drag a model into the thin regions the shift opens up. That’s the concrete reason we lead with the ATT.

6 Variable Structure and Correlations

For context, Figure 6 scatters every variable against every other (one point per sample, the trajectory mean), and Figure 7 boils that down to correlations — Pearson (linear) next to Spearman (rank). The takeaway: **no single variable drives CCN**. The strongest is solar radiation (SWGDN, +0.33), then cloud water, low cloud, and wind speed (all ≈ -0.25) — more particles on sunny, calm, low-cloud days. Pearson and Spearman look almost identical, so these one-to-one relationships are nearly straight-line; the real nonlinearity is in how the variables combine. And CO, the variable that most separates train from test, barely correlates with CCN on its own — it marks *when* smoky air shows up, not how much CCN there is.

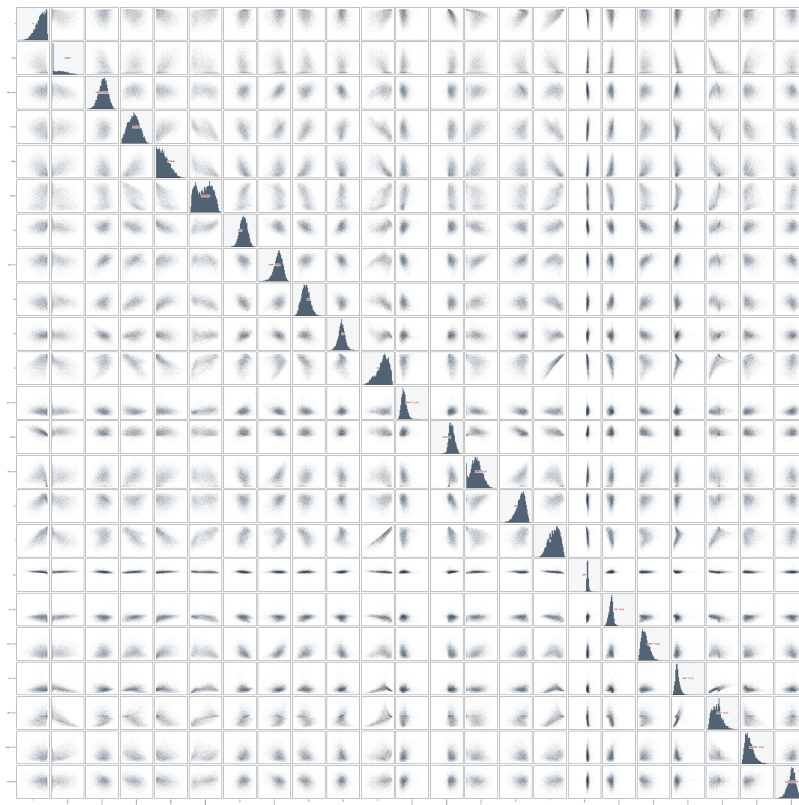


Figure 6: Pairwise scatter matrix of all 22 channels + $\log_{10}(\text{CCN})$ (bottom-right), 6000 samples, per-sample trajectory means. Bottom row / right column = each variable vs. the outcome. Down-scaled; full-resolution version available separately.

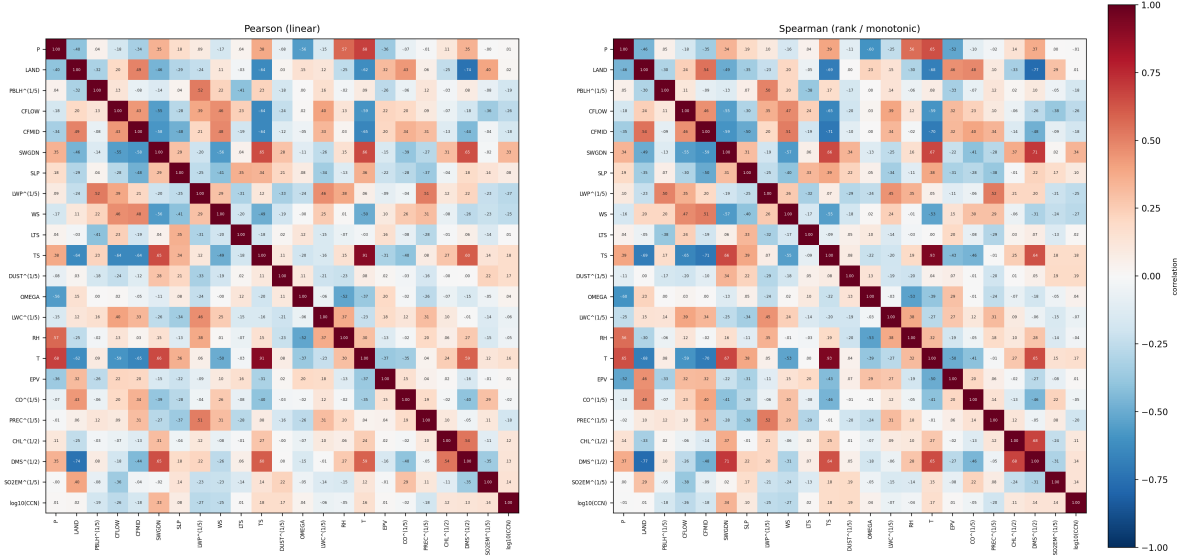


Figure 7: Correlation of all 23 variables over 20,000 samples (per-sample means). Left = Pearson (linear), right = Spearman (rank). Their near-identity indicates the marginal relationships are close to monotone-linear.

7 Distributional Calibration Across the Sweep (PIT)

Coverage (Section 3) is one calibration check; PIT is the sharper one. For each point, the PIT is the fraction of the model’s predicted samples that fall below the true CCN. If the model’s distribution is right these come out uniform, so the histogram should be **flat** — a **U-shape** means the intervals are too narrow (overconfident), a dome means too wide. Across all 48 configs (Figure 8), most are mildly U-shaped: the models are a touch overconfident (the same under-dispersion from Section 3), but none are badly off and the best ones are nearly flat.

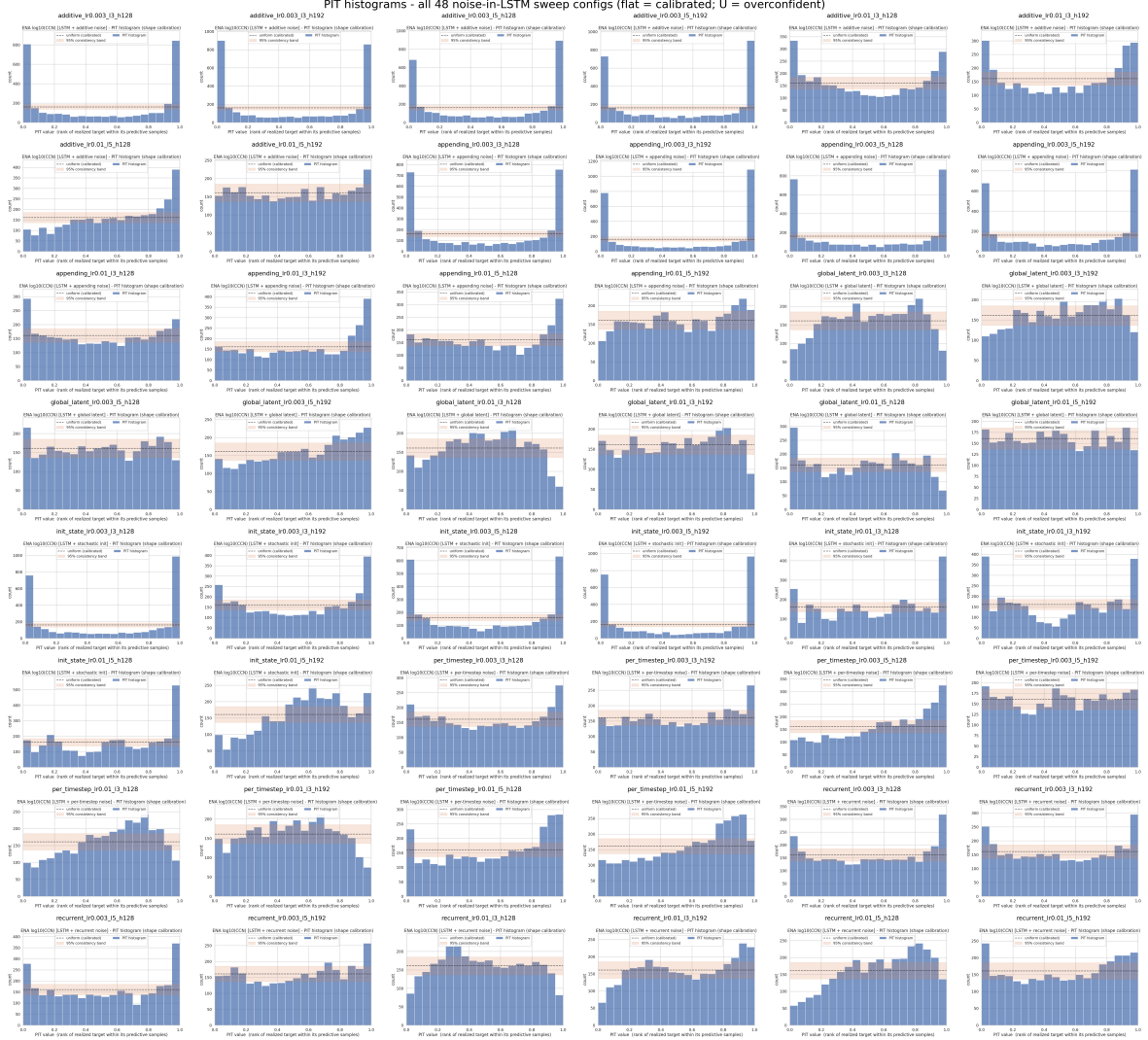


Figure 8: PIT histograms for all 48 sweep configurations (architecture $\times$ hyper-parameters), each titled by its config. Flat = calibrated; U-shape = overconfident; dome = underconfident.

8 Causal Findings: The Wildfire Effect (T-learner)

Now the actual science: does wildfire smoke reaching ENA change CCN? We use a **T-learner** — train one model on wildfire trajectories, another on wildfire-free ones — then run *both* on the same held-out wildfire-period weather. The wildfire model gives the real “with wildfire” CCN; the clean model gives the counterfactual “what it would have been without.” The gap between them is the wildfire effect — an **ATT** (effect on the treated).

Result. Wildfire raises CCN by roughly 15–20% during wildfire periods ($\approx +40$ particles cm^{-3}); the whole distribution shifts up with a fatter high tail, so extreme-CCN episodes become more common. The finding holds across three stochastic-generator variants (Figure 9), each estimating between +10% and +24%.

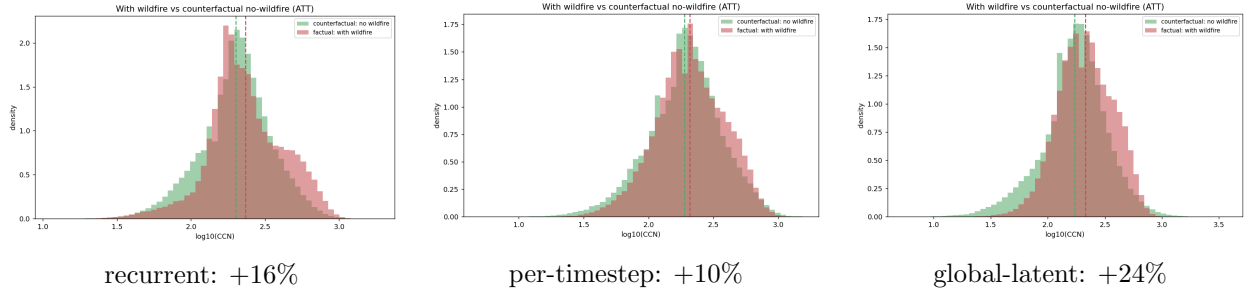


Figure 9: CCN during wildfire periods (red, wildfire model) vs. what it would have been without wildfire (green, clean model), both evaluated at the same held-out wildfire-period weather. Every generator shifts the distribution up (dashed = means) with a fatter high tail. Horizontal axis is a log scale.

The three generators all make the LSTM output a *distribution* by injecting Gaussian noise; they differ in where it enters:

- **Per-timestep:** a noise vector concatenated to the weather inputs, redrawn fresh and independent at every timestep.
- **Global-latent:** the same, except the noise vector is drawn once per trajectory and held fixed across all timesteps (one shared latent code).
- **Recurrent:** unlike the other two, the noise is *added* to the LSTM’s hidden state (not concatenated to the inputs) at each step, so it accumulates through the sequence.

A few caveats. This is the effect during *real* wildfire periods, and it assumes the weather variables capture everything that differs between wildfire and clean air (ignorability). We don’t have formal error bars yet — the spread is +10% to +24% across generators. But because it’s an **ATT measured where both models have data** (overlap holds, per the shift section), it stands on firmer ground than an ATE/ATC that would force the data-poor wildfire model to extrapolate.